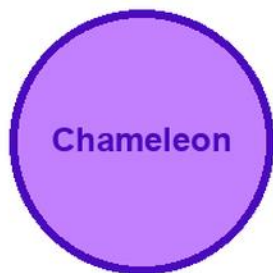


KITABU QUEST

S5

CHEMISTRY



Cover scene

students doing safe chemistry work with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

5

Title Page

KITABU QUEST S5 Chemistry

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S5 learners study Chemistry one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use safe observations, simple models, data tables, and evidence-based explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Alkanes
2. Alkenes and alkynes
3. Halogenoalkanes (alkyl halides)
4. Alcohols and Ethers
5. Carbonyl compounds
6. Carboxylic acids and acyl halides
7. Esters, acid anhydrides, amides and nitriles
8. Amines and amino acids
9. Phase diagrams
10. Solutions and titration
11. Conductivity of solutions
12. Electrolysis

As you finish each topic, return to the success criteria and correct one answer before moving on.

Alkanes: Lesson Opener

Learning focus: Alkanes

Country link: a safe Rwanda observation, model, data table, or investigation for Alkanes.

By the end of this topic, you should be able to:

- the relate the physical and chemical properties of the alkanes to the preparation methods, uses and isomerism
- name straight chain alkanes up to carbon -20
- define homologous series
- be aware of the dangers associated with combustion reactions of the alkanes
- describe and explain the trend in physical properties of homologous series of alkanes
- write the structural formula of the alkanes

Inquiry questions:

- How can alkanes help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkanes: Learn

Alkanes teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Alkanes
- concept
- observation
- evidence
- safety
- conclusion

Alkanes: Worked Example

Worked example for Alkanes: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Alkanes: Vocabulary And Study Skill

Use these words and study moves before you practise Alkanes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Alkanes
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Alkanes without a clear example, method, or evidence.

Alkanes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Alkanes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Alkanes: Practice And Correction

Practice for Alkanes:

Main outcome: the relate the physical and chemical properties of the alkanes to the preparation methods, uses and isomerism.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Alkanes: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Alkanes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkanes: Outcome Check

Outcome check for Alkanes: The relate the physical and chemical properties of the alkanes to the preparation methods, uses and isomerism.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkanes: Outcome Check

Outcome check for Alkanes: Name straight chain alkanes up to carbon -20.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkenes and alkynes: Lesson Opener

Learning focus: Alkenes and alkynes

Country link: a safe Rwanda observation, model, data table, or investigation for Alkenes and alkynes.

By the end of this topic, you should be able to:

- the relate the physical and chemical properties of the alkenes and alkynes to their reactivity and uses
- explain the reactivity of alkenes in comparison to alkanes
- explain the existence of geometrical isomerism in the alkenes
- describe the industrial process of preparing the alkenes and alkynes
- apply IUPAC rules to name alkenes and alkynes
- carry out an experiment to prepare and test ethane gas. Outline the mechanisms for electrophilic addition reactions for the alkenes and alkynes

Inquiry questions:

- How can alkenes and alkynes help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkenes and alkynes: Learn

Alkenes and alkynes is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Alkenes and alkynes
- Alkenes
- alkynes
- concept
- observation
- evidence
- safety
- conclusion

Alkenes and alkynes: Worked Example

Investigation model for Alkenes and alkynes: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Alkenes and alkynes: Vocabulary And Study Skill

Use these words and study moves before you practise Alkenes and alkynes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Alkenes and alkynes
- Alkenes
- alkynes
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Alkenes and alkynes without a clear example, method, or evidence.

Alkenes and alkynes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Alkenes and alkynes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Alkenes and alkynes: Practice And Correction

Practice for Alkenes and alkynes:

Main outcome: the relate the physical and chemical properties of the alkenes and alkynes to their reactivity and uses.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Alkenes and alkynes: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Alkenes and alkynes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkenes and alkynes: Outcome Check

Outcome check for Alkenes and alkynes: The relate the physical and chemical properties of the alkenes and alkynes to their reactivity and uses.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkenes and alkynes: Outcome Check

Outcome check for Alkenes and alkynes: Explain the reactivity of alkenes in comparison to alkanes.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Halogenoalkanes (alkyl halides): Lesson Opener

Learning focus: Halogenoalkanes (alkyl halides)

Country link: a safe Rwanda observation, model, data table, or investigation for Halogenoalkanes (alkyl halides).

By the end of this topic, you should be able to:

- the relate the physical and chemical properties of halogenoalkanes to their reactivity and uses
- define halogenoalkanes and the homologous series
- explain the reactivity of halogenoalkanes
- explain the physical properties of halogenoalkanes
- describe the preparation methods for halogenoalkanes
- explain different mechanisms in halogenoalkane

Inquiry questions:

- How can halogenoalkanes (alkyl halides) help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Halogenoalkanes (alkyl halides): Learn

Halogenoalkanes (alkyl halides) is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Halogenoalkanes (alkyl halides)
- Halogenoalkanes
- alkyl
- halides
- concept
- observation
- evidence
- safety

Halogenoalkanes (alkyl halides): Worked Example

Investigation model for Halogenoalkanes (alkyl halides): Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Halogenoalkanes (alkyl halides): Vocabulary And Study Skill

Use these words and study moves before you practise Halogenoalkanes (alkyl halides). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Halogenoalkanes (alkyl halides)
- Halogenoalkanes
- alkyl
- halides
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Halogenoalkanes (alkyl halides) without a clear example, method, or evidence.

Halogenoalkanes (alkyl halides): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Halogenoalkanes (alkyl halides).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Halogenoalkanes (alkyl halides): Practice And Correction

Practice for Halogenoalkanes (alkyl halides):

Main outcome: the relate the physical and chemical properties of halogenoalkanes to their reactivity and uses.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Halogenoalkanes (alkyl halides): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Halogenoalkanes (alkyl halides).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Halogenoalkanes (alkyl halides): Outcome Check

Outcome check for Halogenoalkanes (alkyl halides): The relate the physical and chemical properties of halogenoalkanes to their reactivity and uses.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Halogenoalkanes (alkyl halides): Outcome Check

Outcome check for Halogenoalkanes (alkyl halides): Define halogenoalkanes and the homologous series.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alcohols and Ethers: Lesson Opener

Learning focus: Alcohols and Ethers

Country link: a safe Rwanda observation, model, data table, or investigation for Alcohols and Ethers.

By the end of this topic, you should be able to:

- the compare the physical and chemical properties of alcohols and ethers to their preparation methods, reactivity and uses
- explain isomers in alcohols from C-4 to C-6
- describe the physical properties and uses of alcohols
- explain the mechanism of dehydration of alcohols and the reaction with hydrogen chloride
- recall the steps involved in fermentation process
- describe the physical and chemical properties and preparation methods of ethers

Inquiry questions:

- How can alcohols and ethers help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alcohols and Ethers: Learn

Alcohols and Ethers is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Alcohols and Ethers
- Alcohols
- Ethers
- concept
- observation
- evidence
- safety
- conclusion

Alcohols and Ethers: Worked Example

Investigation model for Alcohols and Ethers: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Alcohols and Ethers: Vocabulary And Study Skill

Use these words and study moves before you practise Alcohols and Ethers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Alcohols and Ethers
- Alcohols
- Ethers
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Alcohols and Ethers without a clear example, method, or evidence.

Alcohols and Ethers: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Alcohols and Ethers.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Alcohols and Ethers: Practice And Correction

Practice for Alcohols and Ethers:

Main outcome: the compare the physical and chemical properties of alcohols and ethers to their preparation methods, reactivity and uses.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Alcohols and Ethers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Alcohols and Ethers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alcohols and Ethers: Outcome Check

Outcome check for Alcohols and Ethers: The compare the physical and chemical properties of alcohols and ethers to their preparation methods, reactivity and uses.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alcohols and Ethers: Outcome Check

Outcome check for Alcohols and Ethers: Explain isomers in alcohols from C-4 to C-6.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carbonyl compounds: Lesson Opener

Learning focus: Carbonyl compounds

Country link: a safe Rwanda observation, model, data table, or investigation for Carbonyl compounds.

By the end of this topic, you should be able to:

- the compare the chemical nature of carbonyl compounds to their reactivity and uses
- describe the reactivity of carbonyl compounds
- state the physical properties of aldehydes and ketones
- describe the preparation reactions of ketones and aldehydes
- explain the mechanisms of nucleophilic addition reactions of carbonyl compounds
- prepare ketones from secondary alcohols by oxidation reaction

Inquiry questions:

- How can carbonyl compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carbonyl compounds: Learn

Carbonyl compounds is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Carbonyl compounds
- Carbonyl
- compounds
- concept
- observation
- evidence
- safety
- conclusion

Carbonyl compounds: Worked Example

Investigation model for Carbonyl compounds: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Carbonyl compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Carbonyl compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Carbonyl compounds
- Carbonyl
- compounds
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Carbonyl compounds without a clear example, method, or evidence.

Carbonyl compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Carbonyl compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Carbonyl compounds: Practice And Correction

Practice for Carbonyl compounds:

Main outcome: the compare the chemical nature of carbonyl compounds to their reactivity and uses.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Carbonyl compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Carbonyl compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carbonyl compounds: Outcome Check

Outcome check for Carbonyl compounds: The compare the chemical nature of carbonyl compounds to their reactivity and uses.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carbonyl compounds: Outcome Check

Outcome check for Carbonyl compounds: Describe the reactivity of carbonyl compounds.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carboxylic acids and acyl halides: Lesson Opener

Learning focus: Carboxylic acids and acyl halides

Country link: a safe Rwanda observation, model, data table, or investigation for Carboxylic acids and acyl halides.

By the end of this topic, you should be able to:

- the compare the chemical nature of the carboxylic acids and acid halides to their reactivity
- explain the physical properties and uses of the carboxylic acids and acyl chlorides
- describe the inductive effect on the acidity of the carboxylic acids
- explain the reactions of the carboxylic acids and acyl chlorides
- apply the IUPAC rules to name different carboxylic acids acyl chlorides
- write the structural formula and isomers of the carboxylic acids

Inquiry questions:

- How can carboxylic acids and acyl halides help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carboxylic acids and acyl halides: Learn

Carboxylic acids and acyl halides teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Carboxylic acids and acyl halides
- Carboxylic
- acids
- acyl
- halides
- concept
- observation
- evidence

Carboxylic acids and acyl halides: Worked Example

Worked example for Carboxylic acids and acyl halides: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Carboxylic acids and acyl halides: Vocabulary And Study Skill

Use these words and study moves before you practise Carboxylic acids and acyl halides. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Carboxylic acids and acyl halides
- Carboxylic
- acids
- acyl
- halides
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Carboxylic acids and acyl halides without a clear example, method, or evidence.

Carboxylic acids and acyl halides: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Carboxylic acids and acyl halides.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Carboxylic acids and acyl halides: Practice And Correction

Practice for Carboxylic acids and acyl halides:

Main outcome: the compare the chemical nature of the carboxylic acids and acid halides to their reactivity.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Carboxylic acids and acyl halides: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Carboxylic acids and acyl halides.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carboxylic acids and acyl halides: Outcome Check

Outcome check for Carboxylic acids and acyl halides: The compare the chemical nature of the carboxylic acids and acid halides to their reactivity.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carboxylic acids and acyl halides: Outcome Check

Outcome check for Carboxylic acids and acyl halides: Explain the physical properties and uses of the carboxylic acids and acyl chlorides.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Esters, acid anhydrides, amides and nitriles: Lesson Opener

Learning focus: Esters, acid anhydrides, amides and nitriles

Country link: a safe Rwanda observation, model, data table, or investigation for Esters, acid anhydrides, amides and nitriles.

By the end of this topic, you should be able to:

- the relate the functional groups of esters, acid anhydrides, amides and nitriles to their reactivity, preparation methods and uses
- describe the chemical properties of esters, acid anhydrides, amides and nitriles
- describe the process of urea manufacture and its uses
- describe the formation of the detergents
- apply IUPAC rules to name esters, acid anhydrides, amides and nitriles
- compare the physical properties of esters to those of alcohols and carboxylic acids

Inquiry questions:

- How can esters, acid anhydrides, amides and nitriles help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Esters, acid anhydrides, amides and nitriles: Learn

Esters, acid anhydrides, amides and nitriles is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Esters, acid anhydrides, amides and nitriles
- Esters
- acid
- anhydrides
- amides
- nitriles
- concept
- observation

Esters, acid anhydrides, amides and nitriles: Worked Example

Worked example for Esters, acid anhydrides, amides and nitriles: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Esters, acid anhydrides, amides and nitriles: Vocabulary And Study Skill

Use these words and study moves before you practise Esters, acid anhydrides, amides and nitriles. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Esters, acid anhydrides, amides and nitriles
- Esters
- acid
- anhydrides
- amides
- nitriles
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Esters, acid anhydrides, amides and nitriles without a clear example, method, or evidence.

Esters, acid anhydrides, amides and nitriles: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Esters, acid anhydrides, amides and nitriles.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Esters, acid anhydrides, amides and nitriles: Practice And Correction

Practice for Esters, acid anhydrides, amides and nitriles:

Main outcome: the relate the functional groups of esters, acid anhydrides, amides and nitriles to their reactivity, preparation methods and uses.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Esters, acid anhydrides, amides and nitriles: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Esters, acid anhydrides, amides and nitriles.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Esters, acid anhydrides, amides and nitriles: Outcome Check

Outcome check for Esters, acid anhydrides, amides and nitriles: The relate the functional groups of esters, acid anhydrides, amides and nitriles to their reactivity, preparation methods and uses.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Esters, acid anhydrides, amides and nitriles: Outcome Check

Outcome check for Esters, acid anhydrides, amides and nitriles: Describe the chemical properties of esters, acid anhydrides, amides and nitriles.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Amines and amino acids: Lesson Opener

Learning focus: Amines and amino acids

Country link: a safe Rwanda observation, model, data table, or investigation for Amines and amino acids.

By the end of this topic, you should be able to:

- the relate the chemical nature of the amines and amino acids to their properties, uses and reactivity
- explain the zwitterion forms in the solution of different pH
- explain the isoelectric point in amino acids
- describe the physical properties and uses of amines
- describe the preparation methods of the amines
- describe the reactions of amino acids and amines with other substances

Inquiry questions:

- How can amines and amino acids help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Amines and amino acids: Learn

Amines and amino acids is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Amines and amino acids
- Amines
- amino
- acids
- concept
- observation
- evidence
- safety

Amines and amino acids: Worked Example

Investigation model for Amines and amino acids: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Amines and amino acids: Vocabulary And Study Skill

Use these words and study moves before you practise Amines and amino acids. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Amines and amino acids
- Amines
- amino
- acids
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Amines and amino acids without a clear example, method, or evidence.

Amines and amino acids: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Amines and amino acids.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Amines and amino acids: Practice And Correction

Practice for Amines and amino acids:

Main outcome: the relate the chemical nature of the amines and amino acids to their properties, uses and reactivity.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Amines and amino acids: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Amines and amino acids.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Amines and amino acids: Outcome Check

Outcome check for Amines and amino acids: The relate the chemical nature of the amines and amino acids to their properties, uses and reactivity.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Amines and amino acids: Outcome Check

Outcome check for Amines and amino acids: Explain the zwitterion forms in the solution of different pH.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Phase diagrams: Lesson Opener

Learning focus: Phase diagrams

Country link: a safe Rwanda observation, model, data table, or investigation for Phase diagrams.

By the end of this topic, you should be able to:

- the interpret the phase diagrams for different compounds
- explain the term phase equilibrium
- explain the effect of change of state on changing pressure and temperature
- define heterogeneous and homogeneous equilibria
- define triple point, critical point, normal boiling and melting points of substances

Inquiry questions:

- How can phase diagrams help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Phase diagrams: Learn

Phase diagrams is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Phase diagrams
- Phase
- diagrams
- concept
- observation
- evidence
- safety

Phase diagrams: Worked Example

Investigation model for Phase diagrams: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Phase diagrams: Vocabulary And Study Skill

Use these words and study moves before you practise Phase diagrams. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Phase diagrams
- Phase
- diagrams
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Phase diagrams without a clear example, method, or evidence.

Phase diagrams: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Phase diagrams.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Phase diagrams: Practice And Correction

Practice for Phase diagrams:

Main outcome: the interpret the phase diagrams for different compounds.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Phase diagrams: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Phase diagrams.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Phase diagrams: Outcome Check

Outcome check for Phase diagrams: The interpret the phase diagrams for different compounds.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Phase diagrams: Outcome Check

Outcome check for Phase diagrams: Explain the term phase equilibrium.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solutions and titration: Lesson Opener

Learning focus: Solutions and titration

Country link: a safe Rwanda observation, model, data table, or investigation for Solutions and titration.

By the end of this topic, you should be able to:

- the prepare standard solutions and use them to determine concentration of other solutions by titration
- define the terms standard solution and primary standard solution
- prepare solutions with different concentrations
- properly use the burettes, pipettes during titration
- develop a team approach and a sense of responsibility in performing the
- definition of standard solution and primary standard solution

Inquiry questions:

- How can solutions and titration help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solutions and titration: Learn

Solutions and titration is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Solutions and titration
- Solutions
- titration
- concept
- observation
- evidence
- safety
- conclusion

Solutions and titration: Worked Example

Investigation model for Solutions and titration: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Solutions and titration: Vocabulary And Study Skill

Use these words and study moves before you practise Solutions and titration. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solutions and titration
- Solutions
- titration
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solutions and titration without a clear example, method, or evidence.

Solutions and titration: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Solutions and titration.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Solutions and titration: Practice And Correction

Practice for Solutions and titration:

Main outcome: the prepare standard solutions and use them to determine concentration of other solutions by titration.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Solutions and titration: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solutions and titration.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solutions and titration: Outcome Check

Outcome check for Solutions and titration: The prepare standard solutions and use them to determine concentration of other solutions by titration.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solutions and titration: Outcome Check

Outcome check for Solutions and titration: Define the terms standard solution and primary standard solution.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Conductivity of solutions: Lesson Opener

Learning focus: Conductivity of solutions

Country link: a safe Rwanda observation, model, data table, or investigation for Conductivity of solutions.

By the end of this topic, you should be able to:

- the explain the effect of different factors on the molar conductivity of different electrolytes and the applications of conductivity measurements
- explain the conductivity of solutions
- state and explain the factors that affect molar conductivity of solutions
- use Kohlrausch's law to calculate the molar conductivity of an electrolyte
- interpret a graph of molar conductivity against concentration for both weak and strong
- develop a team approach and responsibility in performing experiments

Inquiry questions:

- How can conductivity of solutions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Conductivity of solutions: Learn

Conductivity of solutions connects particle number, amount of substance, mass, volume, pressure, and temperature.

Core idea:

- One mole contains Avogadro's number of particles.
- Molar mass links mass and moles: $\text{moles} = \text{mass} / \text{molar mass}$.
- Gas laws describe how pressure, volume, and temperature change when other conditions are controlled.
- Always list known values, formula, substitution, unit, and final answer.

Key words:

- Conductivity of solutions
- Conductivity
- solutions
- concept
- observation
- evidence
- safety
- conclusion

Conductivity of solutions: Worked Example

Worked example for Conductivity of solutions: Find moles in 18 g of water. Use molar mass of water = 18 g/mol.

1. Formula: moles = mass / molar mass.
2. Substitute: moles = 18 g / 18 g/mol.
3. Calculate: 1 mol.
4. Unit: mol.

Answer: 18 g of water is 1 mol.

Conductivity of solutions: Vocabulary And Study Skill

Use these words and study moves before you practise Conductivity of solutions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Conductivity of solutions
- Conductivity
- solutions
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Conductivity of solutions without a clear example, method, or evidence.

Conductivity of solutions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Conductivity of solutions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Conductivity of solutions: Practice And Correction

Practice for Conductivity of solutions:

Main outcome: the explain the effect of different factors on the molar conductivity of different electrolytes and the applications of conductivity measurements.

1. Calculate moles from a given mass and molar mass.
2. State one gas-law relationship using pressure, volume, or temperature.
3. Show formula, substitution, answer, unit, and one reasonableness check.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Conductivity of solutions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Conductivity of solutions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Conductivity of solutions: Outcome Check

Outcome check for Conductivity of solutions: The explain the effect of different factors on the molar conductivity of different electrolytes and the applications of conductivity measurements.

1. Calculate moles using mass / molar mass.
2. Write the formula and unit.
3. State one controlled condition in a gas-law example.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Conductivity of solutions: Outcome Check

Outcome check for Conductivity of solutions: Explain the conductivity of solutions.

1. Calculate moles using mass / molar mass.
2. Write the formula and unit.
3. State one controlled condition in a gas-law example.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolysis: Lesson Opener

Learning focus: Electrolysis

Country link: a safe Rwanda observation, model, data table, or investigation for Electrolysis.

By the end of this topic, you should be able to:

- the predict the products of given electrolytes during electrolysis and work out quantitatively to determine how much is liberated at a given electrode using Faraday's law
- define electrolysis, cathode and anode
- explain the electrolysis of different substances
- state Faraday's law and define Faraday's constant
- describe the industrial applications of the electrolysis
- develop practical experimental skills related to electrolysis, interpret results and draw valid conclusions

Inquiry questions:

- How can electrolysis help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolysis: Learn

Electrolysis is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Electrolysis
- concept
- observation
- evidence
- safety
- conclusion

Electrolysis: Worked Example

Investigation model for Electrolysis: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Electrolysis: Vocabulary And Study Skill

Use these words and study moves before you practise Electrolysis. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electrolysis
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electrolysis without a clear example, method, or evidence.

Electrolysis: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Electrolysis.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electrolysis: Practice And Correction

Practice for Electrolysis:

Main outcome: the predict the products of given electrolytes during electrolysis and work out quantitatively to determine how much is liberated at a given electrode using Faraday's law.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electrolysis: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electrolysis.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolysis: Outcome Check

Outcome check for Electrolysis: The predict the products of given electrolytes during electrolysis and work out quantitatively to determine how much is liberated at a given electrode using Faraday's law.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolysis: Outcome Check

Outcome check for Electrolysis: Define electrolysis, cathode and anode.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Enthalpy change of reactions: Lesson Opener

Learning focus: Enthalpy change of reactions

Country link: a safe Rwanda observation, model, data table, or investigation for Enthalpy change of reactions.

By the end of this topic, you should be able to:

- the design an experimental procedure to verify the enthalpy changes in a chemical reaction
- define heat of reaction, standard enthalpy change of combustion, enthalpy of neutralisation, enthalpy of solution, enthalpy of hydration and lattice enthalpy
- describe an experimental procedure in determination of heat of combustion
- explain the relationship between quantity of heat produced and mass of substance in a combustion reaction
- develop practical experimental skills about enthalpy changes of reactions, interpreting results and drawing valid conclusions

Inquiry questions:

- How can enthalpy change of reactions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Enthalpy change of reactions: Learn

Enthalpy change of reactions is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Enthalpy change of reactions
- Enthalpy
- change
- reactions
- concept
- observation
- evidence
- safety

Enthalpy change of reactions: Worked Example

Investigation model for Enthalpy change of reactions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Enthalpy change of reactions: Vocabulary And Study Skill

Use these words and study moves before you practise Enthalpy change of reactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Enthalpy change of reactions
- Enthalpy
- change
- reactions
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Enthalpy change of reactions without a clear example, method, or evidence.

Enthalpy change of reactions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Enthalpy change of reactions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Enthalpy change of reactions: Practice And Correction

Practice for Enthalpy change of reactions:

Main outcome: the design an experimental procedure to verify the enthalpy changes in a chemical reaction.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Enthalpy change of reactions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Enthalpy change of reactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Enthalpy change of reactions: Outcome Check

Outcome check for Enthalpy change of reactions: The design an experimental procedure to verify the enthalpy changes in a chemical reaction.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Enthalpy change of reactions: Outcome Check

Outcome check for Enthalpy change of reactions: Define heat of reaction, standard enthalpy change of combustion, enthalpy of neutralisation, enthalpy of solution, enthalpy of hydration and lattice enthalpy.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Entropy and free energy: Lesson Opener

Learning focus: Entropy and free energy

Country link: a safe Rwanda observation, model, data table, or investigation for Entropy and free energy.

By the end of this topic, you should be able to:

- the predict the feasibility of chemical reactions
- explain the term entropy
- state the second law of thermodynamics
- state, if the value of free energy for a reaction will be positive or negative
- relate the entropy changes to the changes in degree of disorder
- predict the spontaneity of reactions using the Gibbs free energy

Inquiry questions:

- How can entropy and free energy help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Entropy and free energy: Learn

Entropy and free energy connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Entropy and free energy
- Entropy
- free
- energy
- concept
- observation
- evidence
- safety

Entropy and free energy: Worked Example

Worked example for Entropy and free energy: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Entropy and free energy: Vocabulary And Study Skill

Use these words and study moves before you practise Entropy and free energy. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Entropy and free energy
- Entropy
- free
- energy
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Entropy and free energy without a clear example, method, or evidence.

Entropy and free energy: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Entropy and free energy.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Entropy and free energy: Practice And Correction

Practice for Entropy and free energy:

Main outcome: the predict the feasibility of chemical reactions.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Entropy and free energy: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Entropy and free energy.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Entropy and free energy: Outcome Check

Outcome check for Entropy and free energy: The predict the feasibility of chemical reactions.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Entropy and free energy: Outcome Check

Outcome check for Entropy and free energy: Explain the term entropy.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Alkanes: Review Clinic 1

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Alkanes that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Alkenes and alkynes: Review Clinic 2

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Alkenes and alkynes that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Halogenoalkanes (alkyl halides): Review Clinic 3

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Halogenoalkanes (alkyl halides) that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Alcohols and Ethers: Review Clinic 4

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Alcohols and Ethers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Chemistry.

- Alkanes: Explain this term using an example from the book, your school, or your community.
- Alkenes and alkynes: Explain this term using an example from the book, your school, or your community.
- Halogenoalkanes (alkyl halides): Explain this term using an example from the book, your school, or your community.
- Alcohols and Ethers: Explain this term using an example from the book, your school, or your community.
- Carbonyl compounds: Explain this term using an example from the book, your school, or your community.
- Carboxylic acids and acyl halides: Explain this term using an example from the book, your school, or your community.
- Esters, acid anhydrides, amides and nitriles: Explain this term using an example from the book, your school, or your community.
- Amines and amino acids: Explain this term using an example from the book, your school, or your community.
- Phase diagrams: Explain this term using an example from the book, your school, or your community.
- Solutions and titration: Explain this term using an example from the book, your school, or your community.
- Conductivity of solutions: Explain this term using an example from the book, your school, or your community.
- Electrolysis: Explain this term using an example from the book, your school, or your community.
- Enthalpy change of reactions: Explain this term using an example from the book, your school, or your community.
- Entropy and free energy: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Chemistry answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Chemistry that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.